THRIVING SCHOLARS

TMUA Mathematical Reasoning Practice

Practice Set
No calculator

10 challenging questions with worked solutions • 2026 edition

About this set: A focused practice set to develop accurate calculation, efficient problem-solving and mathematical reasoning. A Thriving Scholars practice resource; not an official TMUA paper or a prediction of exam content.

Suggested use

Allow about **35–40 minutes**. If you have made no meaningful progress after roughly 2 minutes, flag the question and move on. Choose one answer, A–E, for each question.

1 A real number x, with $x \geq 1$, satisfies

$$\sqrt{(x+1)} + \sqrt{(x-1)} = \sqrt{(3x+1)}.$$

What is the value of $(x^2 + 1)/x$?

A 29/15

B 31/15

C 34/15

D 37/15

E 41/15

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2 The quadratic $x^2 - (k+1)x + 2k + 2 = 0$ has two distinct positive roots. The larger root is exactly twice the smaller root.

What is k?

A −1

B 2

C 5

D 8

E 10

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3 A circle has centre O and radius 5. A chord AB has length 8. The tangents to the circle at A and B meet at T.

What is the area of triangle ATB?

A 16 **B** 20

C 64/3 **D** 24

E 80/3

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4 The real numbers x satisfy

$$2 \sin 2x \cos x = \cos x,$$

where $0 < x < 2\pi$ and $\sin x > 0$. What is the sum of all such values of x?

A $\pi/2$ **B** $2\pi/3$

C $5\pi/6$ **D** π

E $7\pi/6$

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5 Let $f(x) = x^3 - 3x$. How many distinct real solutions does

$$f(f(x)) = 2$$

have?

A 3 **B** 4

C 5 **D** 6

E 7

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6 A five-digit number is formed using five different digits chosen from 1, 2, 3, 4, 5, 6, 7. The number is even, and the digits 1 and 2 are not adjacent.

How many such numbers are possible?

A 720 **B** 792

C 840 **D** 876

E 912

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7 A circle passes through the points $A = (0, 0)$, $B = (8, 0)$, and $C = (2, 6)$. The tangent to the circle at C meets the x-axis at T.

What is AT?

A 6

B 8

C 10

D 12

E 14

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8 Let n be an integer. Consider the statements

P: 24 divides $n^2 - 1$.
Q: n is odd and n is not divisible by 3.

Which statement is correct?

A P is necessary but not sufficient for Q.

B P is sufficient but not necessary for Q.

C P is necessary and sufficient for Q.

D P is neither necessary nor sufficient for Q.

E P and Q cannot both be true.

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9 A student claims: "If real numbers a and b have both $a + b$ rational and ab rational, then a and b must both be rational."

Which pair provides a counterexample to the claim?

A $a = \sqrt{2}$, $b = \sqrt{8}$

B $a = \sqrt{2}$, $b = -\sqrt{2}$

C $a = \sqrt{2}$, $b = 1/\sqrt{2}$

D $a = \sqrt{2}$, $b = 2 - \sqrt{2}$

E $a = \sqrt{3}$, $b = -\sqrt{2}$

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10 The curve $y = x^3 - 3x$ and the line $y = mx + 2$ meet at exactly two distinct points.

What is m?

A -6 **B** -3

C 0 **D** 3

E 6

1	2	3	4	5
____	____	____	____	____
6	**7**	**8**	**9**	**10**
____	____	____	____	____

Answers & Solutions

Keep this section at the back

1. Exact algebra

Answer: C — 34/15

Squaring once gives $2x + 2\sqrt{(x^2 - 1)} = 3x + 1$, so $2\sqrt{(x^2 - 1)} = x + 1$. Squaring again: $4(x^2 - 1) = (x + 1)^2$. Hence $3x^2 - 2x - 5 = 0$, giving $x = 5/3$ (the other root is outside the domain). Therefore $x + 1/x = 5/3 + 3/5 = 34/15$.

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2. Roots and parameters

Answer: D — 8

Let the smaller positive root be r; the other is $2r$. Expanding $(x - r)(x - 2r)$ gives $x^2 - 3rx + 2r^2$. Comparing coefficients, $3r = k + 1$ and $2r^2 = 2k + 2$, so $r^2 = k + 1$. Thus $r^2 = 3r$. Since $r > 0$, $r = 3$, so $k + 1 = 9$ and $k = 8$.

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3. Circle geometry

Answer: C — 64/3

Let M be the midpoint of chord AB. Then $AM = 4$ and $OM = 3$ from the 3–4–5 triangle. Since equal tangents give symmetry, O, M, T are collinear. Right triangles OMA and OAT are similar, so $OT/OA = OA/OM$. Hence $OT = 25/3$ and $TM = 25/3 - 3 = 16/3$. Area = ½ × 8 × 16/3 = 64/3.

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4. Trigonometric structure

Answer: D — π

Rearranging and factorising gives $\cos x(2 \sin 2x - 1) = 0$. Since $\sin x > 0$, the permitted interval is $0 < x < \pi$. The cosine factor gives $x = \pi/2$. The other factor gives $2x = \pi/6$ or $5\pi/6$, so $x = \pi/12$ or $5\pi/12$. Their sum is π. No trigonometric addition formula is needed.

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5. Composite functions

Answer: C — 5

Put $y = f(x)$. Then $f(y) = 2$ gives $y^3 - 3y - 2 = 0 = (y - 2)(y + 1)^2$. So $f(x)$ is either 2 or −1.

$f(x) = 2$ gives $(x - 2)(x + 1)^2 = 0$: two distinct roots. For $f(x) = -1$, $x^3 - 3x + 1 = 0$. Write $g(x) = x^3 - 3x + 1$. Its derivative is $g'(x) = 3x^2 - 3$: g increases for $x < -1$, decreases for $-1 < x < 1$, and increases for $x > 1$. Also $g(-1) = 3$ and $g(1) = -1$. Together with the cubic's end behaviour, this gives exactly one root in each of these three intervals. None can also satisfy $f(x) = 2$. Total: 2 + 3 = 5.

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6. Counting with cases

Answer: D — 876

The final digit is 2, 4 or 6.

If the final digit is 2: there are $6 \times 5 \times 4 \times 3 = 360$ arrangements for the first four positions. The bad cases have 1 immediately before the final 2, giving $5 \times 4 \times 3 = 60$. So this case contributes 300.

If the final digit is 4 or 6: for either choice there are again 360 arrangements. To count bad arrangements, place 1 and 2 as an adjacent block among the first four: choose the other two digits in $(4 \times 3)/2 = 6$ ways, order the three objects in $3 \times 2 \times 1 = 6$ ways, and order 1,2 inside the block in 2 ways. This gives $6 \times 6 \times 2 = 72$ bad arrangements. So each of these two final digits contributes $360 - 72 = 288$.

Total = 300 + 288 + 288 = 876.

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7. Coordinate geometry + tangent

Answer: C — 10

Write the circle as $x^2 + y^2 + Dx + Ey = 0$. Using $B = (8,0)$ gives $D = -8$. Using $C = (2,6)$ then gives $E = -4$. So the centre is (4,2).

The radius from (4,2) to (2,6) has slope −2, hence the tangent at C has slope 1/2. Its equation is $y - 6 = \frac{1}{2}(x - 2)$. Setting $y = 0$ gives $x = -10$. Thus T = (−10,0), so $AT = 10$.

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8. Necessary and sufficient

Answer: C — P is necessary and sufficient for Q

If Q holds, $n - 1$ and $n + 1$ are consecutive even integers: one is divisible by 4 and the other by 2. Their product $n^2 - 1$ is therefore divisible by 8. Among the three consecutive integers $n - 1$, n, $n + 1$, one is divisible by 3. Since n is not, the product $(n - 1)(n + 1)$ is divisible by 3. Because 8 and 3 have no common factor greater than 1, the product is divisible by 24. Thus Q implies P.

Conversely, if n were even, $n^2 - 1$ would be odd. If n were divisible by 3, $n^2 - 1$ would not be divisible by 3. Either case contradicts P. Thus P implies Q as well.

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9. Counterexample reasoning

Answer: B — (√2, −√2)

Here $a + b = 0$ and $ab = -2$, both rational, while neither √2 nor −√2 is rational. One valid counterexample is enough to disprove the universal claim. Option E does not work: its product is −√6, which is irrational.

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10. Repeated-root / intersection reasoning

Answer: C — 0

Intersections satisfy $x^3 - (m + 3)x - 2 = 0$. Exactly two distinct intersections means this cubic has two distinct real roots, with one repeated. Write it as $(x - t)^2(x - u)$. Comparing the coefficients of x^2 gives $u = -2t$. Expanding now gives $x^3 - 3t^2x + 2t^3$.

Comparing constants gives $2t^3 = -2$, so $t = -1$. Comparing the coefficients of x gives $m + 3 = 3t^2 = 3$, hence $m = 0$. The factorisation $(x - 2)(x + 1)^2$ confirms exactly two distinct intersections.
